

## Week 5

### Money Markets

Bullish investors pull \$105bn from US money market funds in four weeks

"Fear of missing out" tempts companies and portfolio managers into riskier bets



---

---

---

---

---

---

---

#### Lecture on the Money Markets

Readings: Burton, M., Nesiba, R., and Brown, B. 2015. An introduction to Financial markets and Institutions. Second Edition. Routledge, Taylor & Francis Group. London & New York, chapter 11

Kacperczyk, M., Schnabl, .2013. HOW SAFE ARE MONEY MARKET FUNDS?. The Quarterly Journal of Economics , Vol. 128, No. 3.

---

---

---

---

---

---

---

At the end of 2021, Apple reported \$64 billion in short-term securities and cash equivalents on its balance sheet. Apple does not keep this in its local bank. But where? This is the topic of the week —**Money Markets**

In this lecture we will discuss the following:

- What the money market is and how it is used by various participants
- The main instruments traded within the money market
- Recent trends in money market instruments

---

---

---

---

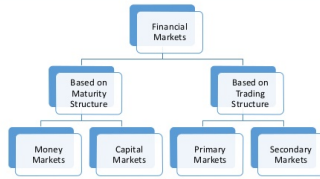
---

---

---

**Recap:**

**Classification of Financial Markets**



Financial Markets are categorized in multiple ways. The most common method adopted is on the basis of maturity and trading. On these 2 criteria, markets are classified under 4 segments. But overall Markets are classified under 8 segments, which is shown in the ensuing slide

---

---

---

---

---

---

---

---

**Recap:**

- Issued in large denominations
    - \$1 million or more
  - Have short maturities
    - One day to one year
    - Typically less than three months
  - Highly liquid; Low default risk
    - Active resale market
  - Market not located at one geographic location
- 
- small changes in interest rates can cause large changes in the prices of these instruments
    - the longer the term to maturity, the greater the change in price for any change in the interest rate

5

---

---

---

---

---

---

---

---

**Economic Role of Money Market (MM)**

- The money market is a market for liquidity.
  - Liquidity is stored in MM by buying MM securities.
  - Liquidity is bought in MM by selling securities (borrowing).
- Provides a place for Central bank's reserve transactions
  - (open market operations).
- Indicator of economic conditions.

6

---

---

---

---

---

---

---

---

Money Market Participants

| Participant                            | How they participate                                                                                                                                        |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Treasury Department                    | Sells Treasury securities to fund the national debt                                                                                                         |
| Central Bank                           | Buys and sells Treasury securities as a way of controlling interest rates                                                                                   |
| Commercial banks                       | Buy Treasury securities; sell certificates of deposit and make short-term loans; offer individual investors accounts that invest in money market securities |
| Companies                              | Buy and sell various short-term securities (commercial paper) as a regular part of their cash management                                                    |
| Investment companies (brokerage firms) | Trade on behalf of commercial accounts                                                                                                                      |
| Insurance companies, pension funds     | Maintain liquidity needed to meet unexpected demands                                                                                                        |
| Money market mutual funds              | Allow small investors to participate in the money market by aggregating their funds to invest in large-denomination money market securities                 |

---

---

---

---

---

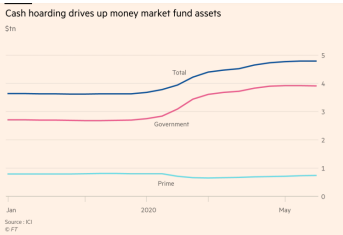
---

---

---

Characteristics of Money Market Instruments

- Low default risk.
- Short maturity.
- High liquidity.
- High marketability.



---

---

---

---

---

---

---

---

Money market funds hit flow record

Weekly mutual and exchange traded money market fund flows (\$bn)



---

---

---

---

---

---

---

---

## Money Market Instruments

- Treasury bills
  - Sold on discount basis.
  - Maturities up to one year.
  - Considered of low default risk.



Source: Bloomberg

10

---

---

---

---

---

---

---

---

## Treasury Bills (cont)

- Pricing Treasury Bills
  - Treasury bills are priced on a bank **discount rate** basis, a traditional yield calculation.

If we need to calculate the yield rate (bond equivalent rate or bey) for a 12-month £100 UK Treasury bill which was sold at a discount of 2 per cent, i.e. at £98, then we recognise that the investment made is £98 and we gain £2 when it is redeemed in one year. Thus we know that the true rate of return is slightly over 2 per cent. Given that there is a full year to maturity (assuming the day-count convention is a 365-day year), we know this is going to be the annual rate:

$$\text{bey} = \frac{\text{Face Value} - \text{Purchase Price}}{\text{Purchase Price}} \times \frac{\text{Days in Year}}{\text{Days to Maturity}} \times 100 = \text{annual rate}$$

$$\text{bey} = \frac{100 - 98}{98} \times \frac{365}{365} \times 100 = 2.04\%$$

---

---

---

---

---

---

---

---

## Money Market Instruments (cont)

- Commercial paper
  - short- term, large denomination, **unsecured** promissory notes issued by the most creditworthy corporations as an alternative to bank borrowing.

12

---

---

---

---

---

---

---

---

### Commercial paper

- Unsecured short-term instrument of debt, issued primarily by corporations
- Promises to the holder a **sum of money to be paid in a set number of days** – it is a promissory note
- Buyers include other corporations, insurance companies, pension funds, governments and banks
- Average maturity of about 40 days; the normal range is 30–90 days, but can be as long as 270 days
- Normally issued at a **discount**.

---

---

---

---

---

---

---

### Commercial paper (cont)

- Discount rate (and yield) higher than for Treasury bills
- Available only to the most respected corporations
- Some companies employ in-house teams to do the selling, other issuers ask dealers
- Can be done in multiple currencies
- Extendable commercial paper (rolling over debt).

---

---

---

---

---

---

---

### Commercial Paper Issuance in the UK



A feature of the 2008 Credit Crunch was the unwillingness of the banks to lend long-term. This led to a spike in CP issuance, as companies turned to short-term sources of finance.

A similar spike happened at the beginning of the pandemic.

---

---

---

---

---

---

---

Commercial paper (cont)

- Because its maturity is less than 270 days, commercial paper does not have to be registered with the Securities and Exchange Commission
- Largest US-based issuers - General Electric Co, Citigroup Inc, Bank of America Corp and Morgan Stanley.



---

---

---

---

---

---

---

---

A dealer buys \$2 million worth of Eurodollar commercial paper from a borrowing company with a 60-day maturity at a discounted price of \$1,994,874. For Eurodollar (and most other Eurocurrency deals) interest is calculated using the 360-day count convention.

The discount yield (using an actual/360 day count) is:

The yield (using an actual/365 day count) is:

$$\text{bey} = \frac{(\text{Face Value} - \text{Purchase Price})}{\text{Purchase Price}} \times \frac{\text{Days in Year}}{\text{Days to Maturity}} \times 100$$
$$\text{bey} = \frac{(2,000,000 - 1,994,874)}{1,994,874} \times \frac{365}{60} \times 100 = 1.5632\% \text{ (annual rate)}$$

---

---

---

---

---

---

---

---

| Moody's   |            | Standard & Poor's |            | Fitch Ratings |            |                    |                             |                    |
|-----------|------------|-------------------|------------|---------------|------------|--------------------|-----------------------------|--------------------|
| Long term | Short term | Long term         | Short term | Long term     | Short term |                    |                             |                    |
| Aaa       | P-1        | AAA               | A-1+       | AAA           | F1+        | Prime              | Investment grade securities |                    |
| Aa1       |            | Aa+               |            | Aa+           |            | High grade         |                             |                    |
| Aa2       |            | Aa                |            | Aa            |            |                    |                             |                    |
| Aa3       |            | A+                |            | A+            |            | Upper medium grade |                             |                    |
| A1        | P-2        | A-                | A-1        | A-            | F1         |                    |                             |                    |
| A2        |            | A                 |            | A             |            |                    |                             | Lower medium grade |
| A3        |            | A-                |            | A-            |            |                    |                             |                    |
| Baa1      |            | P-3               |            | BBB+          | A-3        | BBB+               |                             | F2                 |
| Baa2      | BBB        |                   | BBB        |               |            |                    |                             |                    |
| Baa3      | BBB-       |                   | BBB-       |               |            |                    |                             |                    |

Credit rating systems

---

---

---

---

---

---

---

---

|     |           |      |   |     |   |                                               |                                                     |
|-----|-----------|------|---|-----|---|-----------------------------------------------|-----------------------------------------------------|
| Ba1 |           | BB+  |   | BB+ |   | Somewhat speculative                          |                                                     |
| Ba2 |           | BB   |   | BB  |   | Speculative                                   |                                                     |
| Ba3 |           | BB-  | B | BB- | B |                                               |                                                     |
| B1  |           | B+   |   | B+  |   | Highly speculative                            |                                                     |
| B2  |           | B    |   | B   |   |                                               |                                                     |
| B3  |           | B-   |   | B-  |   |                                               |                                                     |
| Caa | Not Prime | CCC+ |   |     |   | Substantial risks                             | Non-investment grade, high-yield or junk securities |
| Ca  |           | CCC  | C | CCC | C | Extremely speculative                         |                                                     |
| C   |           | CCC- |   |     |   | In default with little prospect for discovery |                                                     |
| /   |           |      |   | DDD |   |                                               |                                                     |
| /   |           | D    | D | DD  | D | In default                                    |                                                     |
| /   |           |      |   | D   |   |                                               |                                                     |

Credit rating systems

---

---

---

---

---

---

---

---

Repurchase agreements (Repos)

- A repo is a way of borrowing for a few days using a sale and repurchase agreement in which securities are sold for cash at an agreed price with a promise to buy back the securities at a specified (higher) price at a future date
- Used very regularly by banks and other financial institutions to borrow money from each other
- This market is also manipulated by central banks to manage their monetary policy
- The term for repos is **usually** between 1 and 14 days

---

---

---

---

---

---

---

---

A high-street bank needs to borrow £6 million for 14 days. It agrees to sell a portfolio of its financial assets, in this case government bonds, to a lender for £6 million. An agreement is drawn up (a repo) by which the bank agrees to repurchase the portfolio 14 days later for £6,001,219.73. The extra sum of £1,219.73 represents the interest on £6 million over 14 days at an annual rate of 0.53 per cent. The calculation is:

Interest = Selling Price × Interest Rate ×  $\frac{\text{Days to Maturity}}{\text{Days in Year}}$

Interest = 6,000,000 ×  $\frac{0.53}{100}$  ×  $\frac{14}{365}$  = £1,219.73

A repo using actual data

---

---

---

---

---

---

---

---

### Certificates of deposit (CDs)

- Issued by banks when funds are deposited with them by other banks, corporations, individuals or investment companies
- Maturities can be any length of time between a week and a year (typically 1–4 months)
- Lots ranging from £50,000 to £500,000 in the UK, or \$100,000 to \$1 million in the US
- **Non-negotiable CDs** must be held by the depositor until the CD reaches maturity
- **Negotiable CD** traded in a secondary market.

---

---

---

---

---

---

---

### Money Market Instruments: The Interbank Market

- Financial institutions borrow/lend short-term funds , usually for a period of one day.
- Used by banks to meet short-term needs
- Banks main alternative to interbank borrowing is to borrow from the Bank of England, the lender of last resort.
- The interbank market is also used by other financial institutions like insurance companies and pension funds that wish to earn a return on their surpluses of ready cash.
- Since the 1980s, the role of the interbank market has broadened considerably.
- The interest rate established in the interbank market – the London Interbank Offered Rate (**LIBOR**) – has been used as the benchmark rate for setting interest rates on a wide variety of loan products – credit card loans, car loans, some mortgages – in Britain and around the world.

---

---

---

---

---

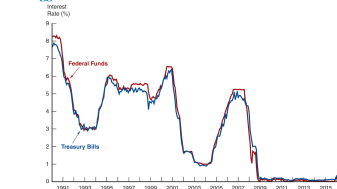
---

---

### The US Interbank Market is called Federal Funds Market

The graph shows federal funds rates and T-bill rates in the US.

Notice that the two rates track fairly closely.  
What does this suggest about the market for T-bills and the market for fed funds?



Source: <http://www.federalreserve.gov/releases/h15/data.htm>

---

---

---

---

---

---

---



**Conclusion**

Money markets remain a crucial part of the financial ecosystem, offering liquidity and stability. However, recent trends indicate a shift towards riskier investments, driven by market optimism. Understanding these shifts is vital for investors and policymakers in maintaining balanced financial strategies.

---

---

---

---

---

---

---